

Excellent

P 16 1.4.1

① $5x^2 - 2x + 9$ 多项式

各项: $5x^2$; $-2x$; 9 (term) ✓

各项系数: $5x^2 \rightarrow 5$; $-2x \rightarrow -2$ (coefficient) ✓

常数项: 9 (constant term) ✓

次数: 3 ✓

② $2a^2 - ab + b = 2 \times (-1)^2 - (-1) \times 3 + 3$ ✓
 $= 2 \times 1 - (-3) + 3$
 $= 2 + 3 + 3$
 $= 8$ ✓

代入求值

③ $7x^2 - 3x + 5 - 2x^2 + 8x - 9 = 5x^2 + 5x - 4$ ✓

④ $4(2x - 3) - 2(3x + 1) = 8x - 12 - 6x - 2$
 $= 2x - 14$ ✓

⑤ $3a - \{2a - [a - 2(a - b)]\} = 3a - \{2a - [a - 2a + 2b]\}$
 $= 3a - \{2a - [-a + 2b]\}$
 $= 3a - \{2a + a - 2b\}$
 $= 3a - \{3a - 2b\}$
 $= 3a - 3a + 2b$
 $= 2b$ ✓

⑥ $2x^2y$ 和 $-5xy^2$ 不是同类项, 因为在 $2x^2y$ 中 x 的指数是 2 和 y 的是 1, 在 $-5xy^2$ 中 x 的指数是 1 和 y 的是 2。它们的指数不同, 所以不是同类项。✓

P 27 A

② $|-6| + 2 - 7 - |-3| = 6 + 2 - 5 - 3$ 绝对值
 $= 6 + 2 - 5 - 3$
 $= 0$ ✓

⑩ $3^4 \times 3^{-2}$

$3^{4-2} = 3^2 = 9$ ✓

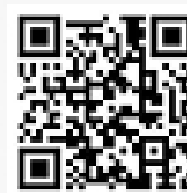
$\frac{x^8}{x^3} = x^5$ ✓

$(2a^2b)^3 = 8a^6b^3$ ✓

⑪ $\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$ ✓

$\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$ ✓

$\sqrt{12a^2} = \sqrt{12} \times \sqrt{a^2} = \sqrt{4 \times 3} \times \sqrt{a^2} = 2\sqrt{3}|a|$



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$$\begin{aligned} (13) \quad 2x^2 - xy + y &= 2x(-2)^2 - (-2) \times 3 + 3 \\ &= 8 + 6 + 3 \\ &= 17 \quad \checkmark \end{aligned}$$

$$(14) \quad 4x^2 - 3x + 7 - 9x^2 + 5x - 2 = -5x^2 + 2x + 5 \quad \checkmark$$

$$\begin{aligned} (15) \quad 3(2x - 5) - (2(x + 4)) &= 6x - 15 - 2x - 8 \\ &= 4x - 23 \quad \checkmark \end{aligned}$$

$$(16) \quad (x + 5)(x - 2) = x^2 + 3x - 10 \quad \checkmark$$

$$(17) \quad (2x - 3)^2 = 4x^2 - 12x + 9 \quad \checkmark$$

P 28 B

$$\begin{aligned} (1) \quad \frac{2a^2 - ab}{a - b} &= \frac{2 \times (-0,5)^2 - (-0,5) \times 3}{-0,5 - 3} = \frac{0,5 + 1,5}{-3,5} = \frac{2}{-3,5} = 2 + \left(-\frac{7}{2}\right) = 2 \times \left(-\frac{2}{7}\right) \\ &= -\frac{4}{7} \quad \checkmark \end{aligned}$$

$$(2) \quad -\frac{9}{4} < -\sqrt{5} < -2,1 < 0 < \sqrt{4} \quad \checkmark$$

$$\begin{aligned} (3) \quad \left(\frac{27}{8}\right)^{\frac{2}{3}} + 16^{\frac{3}{4}} - \sqrt{49} & \quad \left(\frac{27}{8}\right)^{\frac{2}{3}} \rightarrow \left[\left(\frac{3}{2}\right)^3\right]^{\frac{2}{3}} \rightarrow \left(\frac{3}{2}\right)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9} \quad ??? \\ 16^{\frac{3}{4}} &= (2^4)^{\frac{3}{4}} \rightarrow 2^3 = 8 \\ \frac{4}{9} + 8 - 7 &= \sqrt{49} = 7 \end{aligned}$$

教材
1.3.2
1.3.3

$$\hookrightarrow = \frac{4}{9} + \frac{72}{9} - \frac{63}{9} = \frac{13}{9} \quad \checkmark$$

$$\begin{aligned} (7) \quad (2x - 1)(x + 3) - (x - 2)^2 &= 2x^2 + 5x - 3 - (x^2 - 4x + 4) \\ &= 2x^2 + 5x - 3 - x^2 + 4x - 4 \\ &= x^2 + 9x - 7 \quad \checkmark \end{aligned}$$

$$(12) \quad 72500000 = 7,25 \times 10^7$$

$$0,00000408 = 4,08 \times 10^{-6}$$



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P16-1.4.1

P27-A组

(1) 项: $5x^3$	$-2x$	(9) $\rightarrow$ 常数项
系: 5	-2	9
次: 3	1	0

(2) $|-6| + |2-7| - |-3|$

$\Rightarrow 6 + 5 - 3 \Rightarrow 8$

(10) $3^4 \times 3^{-2} = 3^2 = 9$

$\frac{x^8}{x^3} = x^5$

$(2a^2b)^3 = 8a^6b^3$

(2) $2(-1)^2 - (-1)(3) + (3)$

$\Rightarrow 2(1) - (-3) + 3$

$\Rightarrow 2 + 3 + 3 = 8$

(11) $\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$

$\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$

$\sqrt{12a^2} = \sqrt{4 \times 3 \times a^2} = 2a\sqrt{3}$

(3) $7x^2 - 3x + 5 - 2x^2 + 8x - 4$

$\Rightarrow 5x^2 + 5x - 4$

(13) $2(-2)^2 - (-2)(3) + 3$

$= 2(4) - (-6) + 3$

$= 8 + 6 + 3$

$= 17$

(4) $4(2x-3) - 2(3x+1)$

$\Rightarrow 8x - 12 - 6x - 2$

$\Rightarrow 2x - 14$

(14) $4x^2 - 3x + 7 - 9x^2 + 5x - 2$

$= -5x^2 + 2x + 5$

(15) $3(2x-5) - 2(x+4)$

$= 6x - 15 - 2x - 8$

$= 4x - 23$

(5) $3a - \{2a - [a - 2(a-b)]\}$

$\Rightarrow 3a - [2a - (a - 2a + 2b)]$

$\Rightarrow 3a - (2a + a - 2b)$

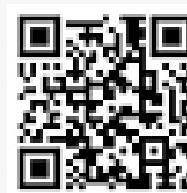
$\Rightarrow 3a - 3a + 2b$

$\Rightarrow 2b$

(16) $(x+5)(x-2) = x^2 - 2x + 5x - 10 = x^2 + 3x - 10$

(17) $(2x-3)^2 = 4x^2 - 12x + 9$

(6) 这两个式子不是同类项因为它们相同的字母不是一样的次数。 $2x^2y$ 里的 x 的次数是2, 但 $-5xy^2$ 里的 x 的次数是1。



Date.

P28-B组

$$(1) \frac{2(-\frac{1}{2})^2 - (-\frac{1}{2})(3)}{(-\frac{1}{2}) - (3)}$$

$$= \frac{2(\frac{1}{4}) - (-\frac{3}{2})}{-\frac{7}{2}}$$

$$= \frac{\frac{1}{2} + \frac{3}{2}}{-\frac{7}{2}}$$

$$= 2 \times (-\frac{2}{7}) = -\frac{4}{7} \checkmark$$

$$(2) -\frac{9}{4} = -\sqrt{\frac{81}{16}} \quad -\sqrt{5} = -\sqrt{\frac{80}{16}}$$

$$-\frac{9}{4}, -\sqrt{5}, -2.1, 0, \sqrt{4} \checkmark$$

$$(3) (\frac{27}{8})^{-2/3} + 16^{3/4} - \sqrt{49}$$

$$= \frac{1}{(\sqrt[3]{\frac{27}{8}})^2} + (\sqrt[4]{16})^3 - 7$$

$$= \frac{1}{(\frac{3}{2})^2} + 2^3 - 7$$

$$= \frac{4}{9} + 8 - 7$$

$$= \frac{4}{9} + 1$$

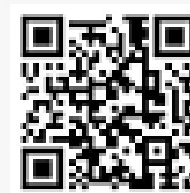
$$= \frac{13}{9} \checkmark$$

$$(7) (2x-1)(x+3) - (x-2)^2$$

$$= 2x^2 + 6x - x - 3 - (x^2 - 4x + 4)$$

$$= 2x^2 + 5x - 3 - (x^2 - 4x + 4)$$

$$= x^2 + 9x - 7 \checkmark$$



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Excellent

① 指出 $5x^3 - 2x + 9$ 的各项、各项系数、常数项和次数。

Terms(各项)	$5x^3$	$-2x$	9
• Coefficient(系数)	5	-2	9
• Degree(次数)	3	1	0

Constant term: 常数项: 9

② 当 $a = -1$, $b = 3$ 时, 求 $2a^2 - ab + b$ 的值。

$$2a^2 - ab + b; a = -1, b = 3$$

$$= 2(-1)^2 - (-1)(3) + 3 = 2 + 3 + 3 = 8$$

③ 合并同类项: $7x^2 - 3x + 5 - 2x^2 + 8x - 9$

$$= 7x^2 - 2x^2 - 3x + 8x + 5 - 9$$

$$= 5x^2 + 5x - 4$$

④ 去括号并化简: $4(2x - 3) - 2(3x + 1)$

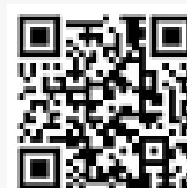
$$= (8x - 12) - (6x + 2) = 8x - 12 - 6x - 2 = 2x - 14$$

⑤ 去括号并化简: $3a - \{2a - [a - 2(a - b)]\}$

$$= 3a - \{2a - [a - (2a - 2b)]\}$$

$$= 3a - [2a - (a - 2a + 2b)]$$

$$= 3a - [2a - (-a + 2b)]$$



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$$= 3a - (2a + a - 2b)$$

$$= 3a - 2a - a + 2b$$

$$= 2b$$

⑥ 判断 $2x^2y$ 与 $-5xy^2$ 是否为同类项, 并说明理由。

Eng. NO, they are not like terms. For terms to be "like terms", they must have the same variables, and the same exponent. Even though they have the same letters (x and y), the exponents are different. In the first term ($2x^2y$), x is squared, and y is to the power of 1. However for the second term ($-5xy^2$), y is being squared, and x is raised to the power of 1.

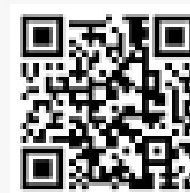
Ch: 不是同类项。要成为“同类项”，它们必须包含完全相同的变量，并且变量的次数也必须完全相同。即使它们有相同的变量 (x 和 y)，但次数不同。在第一项 ($2x^2y$)， x 是平方， y 是 1 次方。然而在第二项 ($-5xy^2$) 中， y 是平方， x 是 1 次方。

P.27 A(2)(10)(11)(13)(14)(15)(16)(17)

(2) 计算: $|-6| + |2-7| - |-3|$

$$= 6 + |-5| - |-3|$$

$$= 6 + 5 - 3 = 8$$



$$(10) \text{ 化简: } 3^4 \times 3^2, \frac{x^8}{x^3}, (2a^2b)^3$$

$$3^4 \times 3^2$$

$$= 3^{4+2} = 3^6 = 9$$

$$\frac{x^8}{x^3}$$

$$= x^{8-3} = x^5$$

$$(2a^2b)^3$$

$$= (2)^3 \times (a^2)^3 \times (b)^3$$

$$= 8a^6b^3$$

$$(11) \text{ 化简: } \sqrt{32}, \sqrt{75}, \sqrt{12a^2}$$

$$\sqrt{32}$$

$$= \sqrt{16 \times 2} = 4\sqrt{2}$$

$$\sqrt{75}$$

$$= \sqrt{25 \times 3} = 5\sqrt{3}$$

$$\sqrt{12a^2}$$

$$= \sqrt{4 \times 3 \times a^2} = 2a\sqrt{3}$$

(13) 当 $x=-2, y=3$ 时, 求 $2x^2 - xy + y$ 的值。

$$= 2(-2)^2 - (3)(-2) + (3)$$

$$= (2)(4) - (-6) + 3$$

$$= 8 + 6 + 3 = 17$$

$$(14) \text{ 合并同类项: } 4x^2 - 3x + 7 - 9x^2 + 5x - 2$$

$$= 4x^2 - 9x^2 - 3x + 5x + 7 - 2$$

$$= -5x^2 + 2x + 5$$

(15) 去括号并化简:

$$3(2x-5) - 2(x+4)$$

$$= (6x-15) - (4x+8)$$

$$= 6x-15-4x-8$$

$$= 2x-23$$

(16) 展开: $(x+5)(x-2)$

$$= x^2 - 2x + 5x - 10$$

$$= x^2 + 3x - 10$$

(17) 展开: $(2x-3)^2$

$$= 4x^2 - 12x + 9$$

P.28 B(1)(2)(3)(7)

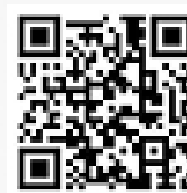
(1) 已知 $a = -\frac{1}{2}, b = 3$,

$$\text{计算 } \frac{2a^2 - ab}{a-b}$$

$$= \frac{2(-\frac{1}{2})^2 - (-\frac{1}{2})(3)}{-\frac{1}{2} - 3}$$

$$= \frac{2(\frac{1}{4}) - (-\frac{3}{2})}{-\frac{7}{2}}$$

$$= -\frac{7}{2}$$



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$$= \frac{\frac{1}{2} + \frac{3}{2}}{-\frac{7}{2}}$$

$$= \frac{\frac{4}{2}}{-\frac{7}{2}}$$

$$= \frac{4}{2} \times \left(-\frac{2}{7}\right)$$

$$= 2 \times \left(-\frac{2}{7}\right)$$

$$= -\frac{4}{7}$$

(2) 按照从小到大的顺序排

列 $-\sqrt{5}, -2.1, -\frac{9}{4}, 0, \sqrt{4}$

答: $-\frac{9}{4}, -\sqrt{5}, -2.1, 0, \sqrt{4}$

(3) 计算并化简:

$$\left(\frac{27}{8}\right)^{-\frac{2}{3}} + 16^{\frac{3}{4}} - \sqrt{49}$$

$$= \left[\left(\frac{3}{2}\right)^3\right]^{-\frac{2}{3}} + \left[(2)^4\right]^{\frac{3}{4}} - 7$$

$$= \left(\frac{3}{2}\right)^{-2} + 2^3 - 7$$

$$= \left(\frac{2}{3}\right)^2 + 8 - 7$$

$$= \frac{4}{9} + 8 - 7$$

$$= \frac{4+72-63}{9} = \frac{13}{9}$$

(7) 展开并化简:

$$(2x-1)(x+3) - (x-2)^2$$

$$= (2x^2 + 6x - x - 3) - (x^2 - 4x + 4)$$

$$= 2x^2 + 5x - 3 - x^2 + 4x - 4$$

$$= x^2 + 9x - 7$$

